

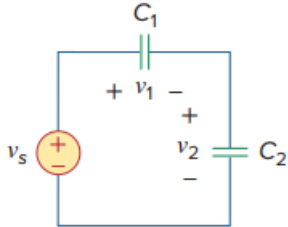
Problem

(a) Show that the voltage-division rule for two capacitors in series as in Fig. 6.59(a) is

$$v_1 = \frac{C_2}{C_1 + C_2} v_s, \quad v_2 = \frac{C_1}{C_1 + C_2} v_s$$

assuming that the initial conditions are zero.

Fig. 6.59(a):

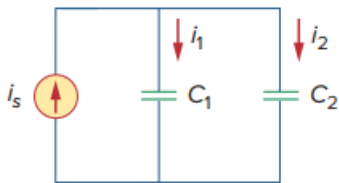


(b) For two capacitors in parallel as in Fig. 6.59(b), show that the current-division rule is

$$i_1 = \frac{C_1}{C_1 + C_2} i_s, \quad i_2 = \frac{C_2}{C_1 + C_2} i_s$$

assuming that the initial conditions are zero.

Fig. 6.59(b):



Step-by-step solution

Step 1 of 3

(a)

Refer to Figure 6.59(a) in the textbook for the series connection of capacitors in a circuit.

Charge enclosed Q in a capacitor is given by,

$$Q = CV$$

Here, C is the capacitance of capacitor. and V is the voltage across the capacitor.

Charge enclosed by the capacitors in the series connection is equal, hence,

$$Q_1 = Q_2$$

Substitute the expression of charges in the equation and solve for the voltage expression.

$$C_1 v_1 = C_2 v_2$$

$$\frac{v_1}{v_2} = \frac{C_2}{C_1}$$

$$v_1 = \frac{C_2 v_2}{C_1} \quad \text{.....(1)}$$

$$v_2 = \frac{C_1 v_1}{C_2} \quad \text{.....(2)}$$

Apply Kirchhoff's voltage law to the circuit.

$$v_s = v_1 + v_2$$

$$= \frac{C_2 v_2}{C_1} + v_2 \quad \left(\text{since, } v_1 = \frac{C_2 v_2}{C_1} \right)$$

$$= v_2 \left(\frac{C_2}{C_1} + 1 \right)$$

$$= v_2 \left(\frac{C_2 + C_1}{C_1} \right)$$

$$v_2 = \left(\frac{C_1}{C_1 + C_2} \right) v_s$$

Step 2 of 3

Apply Kirchhoff's voltage law to the circuit.

$$v_s = v_1 + v_2$$

$$= v_1 + \frac{C_1 v_1}{C_2} \quad \left(\text{since, } v_2 = \frac{C_1 v_1}{C_2} \right)$$

$$= v_1 \left(1 + \frac{C_1}{C_2} \right)$$

$$= v_1 \left(\frac{C_2 + C_1}{C_2} \right)$$

$$v_1 = \left(\frac{C_2}{C_1 + C_2} \right) v_s$$

$$v_1 = \left(\frac{C_2}{C_1 + C_2} \right) v_s$$

Therefore, voltage division rule for two capacitors in series is proved and is,

$$\boxed{v_2 = \left(\frac{C_1}{C_1 + C_2} \right) v_s}, \text{ and } \boxed{v_1 = \left(\frac{C_2}{C_1 + C_2} \right) v_s}.$$

Step 3 of 3

(b)

Refer to Figure 6.59(b) in the textbook for the parallel connection of capacitors in a circuit.

Consider that the voltage drop across the capacitors in parallel connection is equal.

$$v_1 = v_2$$

Substitute the expression of voltages in the equation and solve for the current expression.

$$\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$Q_1 C_2 = Q_2 C_1$$

$$Q_1 = \frac{Q_2 C_1}{C_2}$$

Consider that the total charge enclosed by the capacitors in the parallel connection is equal to the sum of the individual charges.

$$Q_s = Q_1 + Q_2$$

$$= \frac{C_1}{C_2} Q_2 + Q_2 \quad \left(\text{since, } Q_1 = \frac{Q_2 C_1}{C_2} \right)$$

$$= \left(\frac{C_1}{C_2} + 1 \right) Q_2$$

$$= \left(\frac{C_1 + C_2}{C_2} \right) Q_2$$

Therefore,

$$Q_2 = \left(\frac{C_2}{C_1 + C_2} \right) Q_s$$

Similarly,

$$Q_1 = \left(\frac{C_1}{C_1 + C_2} \right) Q_s$$

Therefore, the current-division rules for the parallel connection of two capacitors is proved and

is,
$$Q_2 = \left(\frac{C_2}{C_1 + C_2} \right) Q_s, \text{ and } Q_1 = \left(\frac{C_1}{C_1 + C_2} \right) Q_s$$